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Laminarflame speeds, non-premixed stagnation ignition, and reduced mechanisms in the oxidation of *iso*-octane

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Abstract

Experimental data on the laminar flame speeds of *iso*-octane/air mixtures at atmospheric and elevated pressures were acquired using the counterflow flame and the outwardly expanding flame, while the non-premixed ignition temperatures were determined for an *iso*-octane pool in the stagnation flow of a heated air jet at atmospheric and slightly reduced/elevated pressures. These experimental measurements were compared with calculations based on the mechanisms of Curran et al. and Chaos et al., with the former mechanism systematically and substantially reduced, using directed relation graph and computational singular perturbation, to facilitate the calculation. It was found that the Curran mechanism yielded substantially higher laminar flames speeds as compared to the present experimental results while results from the Chaos mechanism agree well with the present measurements. These trends are in agreement with previous results on the laminar flame speeds for *n*-heptane. Both mechanisms yield acceptable comparison with the observed non-premixed stagnation ignition temperature.
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Keywords: *iso*-octane; Primary reference fuel; Flame speed; Ignition temperature; Reduced mechanism

1. Introduction

In spite of the importance of *iso*-octane and *n*-heptane as the components constituting primary reference fuels (PRFs) for the octane ratings of gasoline fuels, there still exists considerable uncertainty in their respective and joint oxidation mechanisms, as well as disagreement in the reported experimental and computed values of their various fundamental combustion properties. Furthermore, compared to *n*-heptane, the extent of investigation for *iso*-octane has been less extensive, and as such the state of uncertainty is correspondingly more substantial. For example, literature experimental data reveal significant discrepancies in the measured laminar flame speeds, which are of relevance to high-temperature chemistry. Specifically, at 1 atmosphere and 298 K, the measurements of Davis and Law [1] and Huang et al. [2] are nearly identical while those of Kwon et al. [3] are lower for lean mixtures and higher for rich mixtures, often by more than 6 cm/s which is well outside of the reported experimental uncertainty. Discrepancies are also seen at elevated temperatures and pressures [4,5]. Furthermore, there has been only one set of experimental measurements on the diffusive ignition of *iso*-octane, which is of relevance to intermediate-temperature chemistry, involving the ignition of an *iso*-octane pool by an impinging heated air jet [6] and global characterization of the flow system. Thus there exist a critical need

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for well-characterized, highly-accurate fundamental experimental data for *iso*-octane that can be used not only as input to the understanding of more complex combustion phenomena, but also as targets in the development of its oxidation mechanisms. Finally, we also note that while viable mechanisms have recently been developed [7,8], critical comparisons of their performance against high fidelity flame experiments have not been systematically conducted – one reason being the lack of such data as just mentioned.

In view of the above considerations, we have conducted a fairly comprehensive investigation in order to meet these needs. Specifically, we have acquired laminar flame speed data using two entirely different experimental configurations, namely the counterflow flame and the spherically expanding flame, thereby imposing a stringent check on the internal consistency and accuracy of the data. For diffusive ignition, we have employed yet a third experimental set-up, namely the stagnation pool experiment of Ref. [9], but with the flow field locally characterized so that the results are of more general utility. For all these experiments we have preformed the measurement at different pressures whenever feasible, and as such introduced an additional kinetic constraint that allows further scrutiny of the comprehensiveness of the reaction mechanism. Finally, we have reduced a well-established detailed *iso*-octane oxidation mechanism to substantially smaller sizes, using several recently developed reduction algorithms of controlled accuracy. These reduced mechanisms are of value not only for their potential integration in the simulation of more complex combustion phenomena, but their substantially reduced computational demand is essential in our ability to conduct comparisons of the experimental data with the simulated ones, as will be demonstrated later.

2. Determination of laminar flame speeds

2.1. Experimental

Flame speeds were acquired using the counterflow twin-flame technique and the dual-chamber expanding flame technique. The former technique is well established, and its adaption for volatile liquid fuels, as for *iso*-octane, is discussed in Ref. [10] for *n*-heptane. The latter technique allows experimentation in a constant- and high-pressure (up to 30–60 atm) environment [11], and its adaption for volatile liquid fuels is discussed in Ref. [12]. Stretch effects due to the divergent flow field of the former and the expansion of the spherical flame of the latter were subtracted out to yield the stretch-free laminar flame speeds. Note that while linear extrapolation was performed on the counterflow data as it was demonstrated [1] to be adequate and consistent with the accuracy of the data; a nonlinear extrapolation was used for the expanding flames for which the nonlinear effect has been demonstrated to be important for heavier fuels [13]. The experimental data were estimated to have an uncertainty of ±2 cm/s due to data repeatability and data selection for extrapolation.

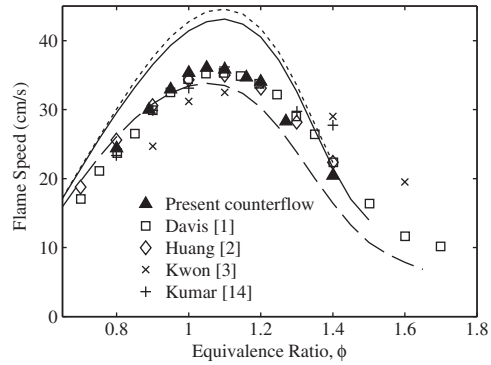

Fig. 1. Laminar flame speed of *iso*-octane/air. 1 atm, $T_u = 298$ K. Dashed-line: Mech-PRF, solid-line: Mech-112, dotted-line: Mech-78.

Figure 1 is a plot of the present counterflow data at 298 K initial temperature against previous measurements. It is seen that, overall, there is very good agreement between the present measurements and those of Davis and Law [1] and of Huang et al. [2]. The results of Kwon et al. [3] are significantly lower and higher on the lean and rich sides, respectively, while those of Kumar et al. [14] appear to deviate substantially from the present result for the richer mixtures.

Figure 2 continues the comparison at 1 atm, but at elevated temperatures, in the range of 350–360 K. Here we first note that the present measurements, obtained independently using the two different configurations of counterflow flame and expanding flame, agree well, hence providing an

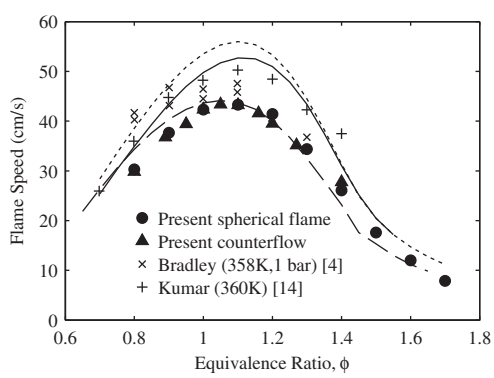

Fig. 2. Laminar flame speed of *iso*-octane/air. 1 atm, $T_u = 353$ K. Dashed-line: Mech-PRF, solid-line: Mech-112, dotted-line: Mech-78.

internal consistency check of the accuracy and meaningfulness of the data. Additionally, the mass flux is nearly identical for the counterflow data shown in Figs. 1 and 2, indicating that the leading order effect of increasing the unburned gas temperature is through the change in the unburned gas density. This further supports the self-consistency of the data.

The measurements of Bradley et al. [4] are substantially different from the present values on the lean side while they are relatively similar on the rich side. To resolve this discrepancy, the stretched flame speed data of Bradley et al. are compared with the present spherical flame data in Fig. 3. The data used for extrapolation are shown in dark color, while the data affected by ignition or confinement were not used for extrapolation and are shown in grey. This figure clearly indicates that a discrepancy exists between the measured stretched downstream flamespeed of Bradley et al. and the present measurements. Therefore, the discrepancy between the reported unstretched laminar flame speeds is due to neither the extrapolation nor data selection; rather the experimental measurements are fundamentally different for the lean experiments. As numerous factors could have contributed to such a difference, we are unable to comment on the source of the discrepancy and as such can only comment on our measurements.

The measurements of Kumar et al. [14] are uniformly higher than the present measurements by approximately 8 cm/s, which is too much to be accounted for by its slightly higher initial temperature, 360 K. This trend was similarly seen for elevated temperature *n*-heptane measurements in that the data of Kumar et al. were consistently higher than measurements by the present authors [10,12] as well as Ji et al. [15].

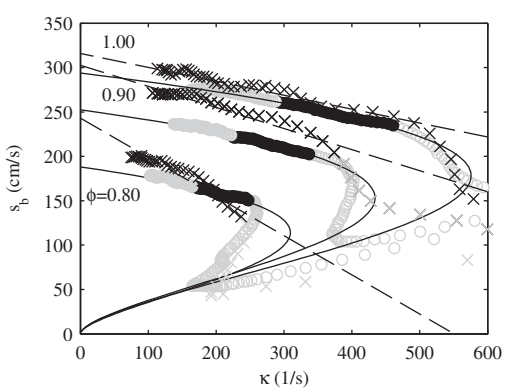

Fig. 3. Experimentally measured stretched downstream flamespeed, s_b, as a function of stretch rate, κ for three equivalence ratios. Crosses: Bradley ($T_u = 358$ K). Circles: present spherical flame. Dark symbols: data used for extrapolation. Solid lines: present nonlinear extrapolation. Dashed-line: Bradley linear extrapolation.

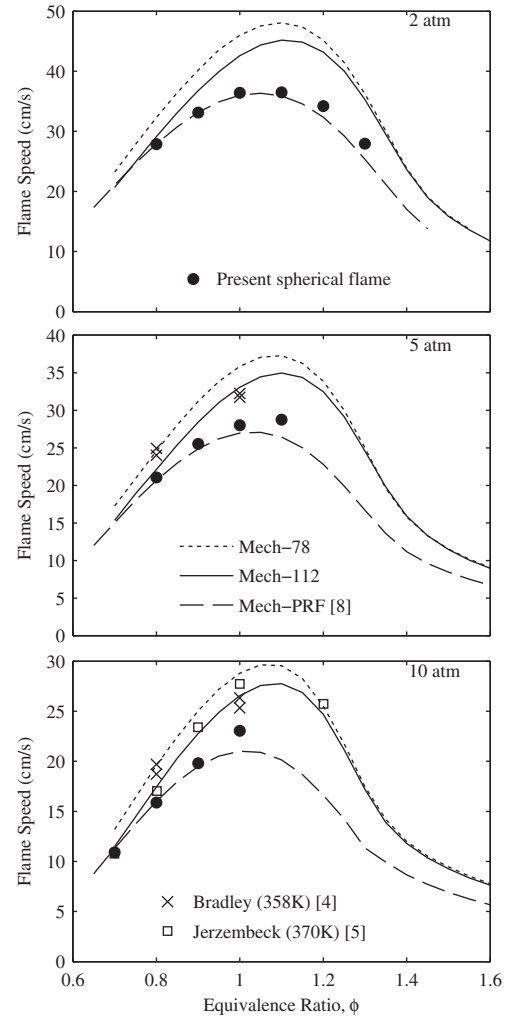

Fig. 4. Laminar flame speed of *iso*-octane/air at various pressures, $T_u = 353$ K.

Figure 4 shows our data at 2, 5, and 10 atm pressures; the upper limit in the equivalence ratio for which data is shown is due to the onset of flamefront instability [16]. It is seen that the measurements of Bradley et al. [4] and by Jerzembeck et al. [5] are consistently higher than the present ones.

2.2. Reaction mechanisms

We have compared our experimental measurements of laminar flame speeds, and non-premixed stagnation-flow ignition temperatures to be reported later, with the predictions of two widely adopted chemical kinetic mechanisms: the high-temperature mechanism of Chaos et al. [8] for the primary reference fuel and the comprehensive mechanism of Curran et al. [7] for *iso*-octane,

which we shall, respectively designate as Mech-PRF and Mech-LLL. Due to the large size of the latter mechanism, simulation of even the simplest combustion phenomenon, namely the 1-D adiabatic planar flame propagation in the doubly infinite domain which yields the laminar flame speed eigenvalue, is extremely computationally demanding. As such, this mechanism was first reduced to a skeletal mechanism consisting of 112 species through time-scale analysis, with minimal loss of accuracy through both stages of reduction. In conducting the reduction, auto-ignition delays, extinction residence times of PSR, and laminar flame speeds were chosen as the target parameters, while conditions over a wide parameter range were covered: pressure from 1 to 40 atm, initial temperature from 600 to 1800 K for auto-ignition, and equivalence ratio from 0.5 to 1.5. Details of the reduction process can be found in e.g. Ref. [17]. Our previous study on *n*-heptane [17] additionally demonstrated that the skeletal mechanism can accurately and comprehensively represent the detailed mechanism. As the reduction process uses a wide range of target parameters, the reduced mechanism is useful for both high-temperature and low-temperature calculations.

2.3. Comparisons

The above experimental results are compared with the calculated laminar flame speeds using Mech-112, Mech-78, and Mech-PRF in Figs. 1, 2 and 4. It is readily seen that calculations using Mech-78 agree well with those obtained with Mech-112, demonstrating the there is minimum loss of accuracy in the second-stage, time-scale reduction. Since Mech-112 is also expected to closely resemble Mech-LLL, we shall consider these three mechanisms as basically equivalent in our assessment of their performance, and quote Mech-LLL as their representative mechanism.

The comparison then shows that while Mech-PRF predicts quite well the present experimental data at all pressures, as well as those of Huang et al. and Davis and Law at 1 atm, Mech-LLL substantially over-predicts them for all situations. The same extent of difference was also observed for the laminar flame speeds of *n*-heptane [10,17] compared to the Lawrence Livermore mechanism, and the difference far exceeds the uncertainty of the experimental data for both *iso*-octane and *n*-heptane, obtained with the counterflow and expanding flame techniques. On the other hand, Mech-LLL does seem to predict well the experimental data of Bradley et al. [4] and Jerzembeck et al. [5].

As an effort to identify the dominant reactions in these two mechanisms that led to the observed differences, we have evaluated a disparity coefficient defined as,

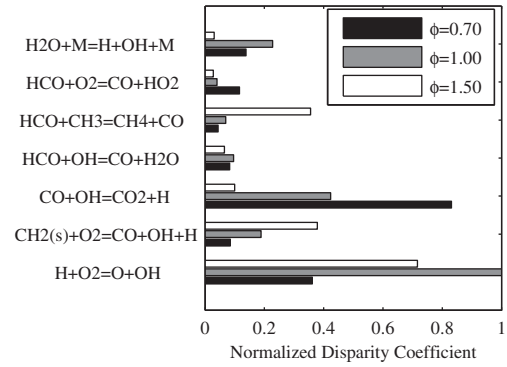

Fig. 5. Normalized disparity coefficient defined by Eq. (1) to determine the reactions which are the source of the discrepancy between Mech-LLL and Mech-PRF. Shown are the reactions with the largest disparity coefficient for atmospheric pressure, $T_u = 353$ K.

$$|(\partial \ln f/\partial \ln k_i)_\phi (1 - k_{i,Mech-PRF}/k_{i,Mech-112})_{T_{ad}(\phi)}| \tag{1}$$

which is the absolute value of the product of the normalized difference in the rates of a reaction, *i*, evaluated by the two mechanisms, and the sensitivity coefficient of that reaction for the mass burning flux *f*, which is the density-weighted laminar flame speed. Extensive calculations were performed for various pressures and equivalence ratios to determine the most important reactions common to all conditions responsible for the observed differences. Figure 5 is a normalized plot of the most significant disparity coefficients showing, perhaps not surprisingly, that the reactions $H + O_2 = O + OH$ and $CO + OH = CO_2 + H$ are the leading candidates. It is beyond the scope of the present study to suggest modification of these rates to improve the mechanism. A further study of the transport properties revealed that they are similar for the two mechanisms and are therefore not responsible for the notable differences in the flame speed predications.

3. Ignition in the stagnation flow

While laminar flame speeds are relevant to the study of high-temperature chemistry, ignition in a strained, diffusive flow field extends the inquiry of chemistry into the lower temperature range. In fact, since the ignition temperatures of hydrocarbons in a strained flow field still exceed say 1000 K because of the constraint of the finite residence time, and since it is conservative to expect that intermediate-temperature chemistry could still influence ignition, a study of the present phenomenon should include chemistry traversing this extended temperature range of interest.

The present experiment involves impinging an electrically heated oxidizer stream onto a liquid fuel pool to induce ignition for a given oxidizer temperature and flow strain rate. This configuration is preferred over that of the counterflow because of the progressive difficulty in generating and maintaining a fuel jet with enough fuel vapor to effect ignition as the fuel becomes more nonvolatile and/or for experiments at higher pressures.

3.1. Experimental and computational specifications

Detailed experimental specifications are given in Ref. [9]. Briefly, the apparatus consists of an upper burner that directs the heated oxidizer jet of temperature up to 1300–1400 K downward to impinge on a pool of liquid fuel, whose level was maintained at a fixed location and its temperature is monitored. An LDV was used to measure the axial flow velocity and hence determine the local strain rate of the flow. The burner was housed in a chamber that allowed experimentation at reduced and elevated pressures. The state of ignition was brought about, and monitored, by fixing the oxygen concentration and the mass flux of the oxidizer and gradually increasing the air temperature.

Computational simulation of ignition with detailed chemistry and transport in laminar flows was conducted by using a modified version of the "Opposed-Flow Diffusion Flames" (OPPDIF) code [18], which computes the steady-state solution for planar diffusion flames in the counterflow configuration. To adapt it to the present stagnation flow, the boundary conditions on the fuel side were modified as discussed in Ref. [19]. The characteristic local strain rate, for both experimental measurements and numerical simulations, was determined by fitting the axial velocity data within the range that roughly spans the ignition kernel in the present stagnation flow. Consequently the strain rate experienced by the flow within this region is a realistic representation of the characteristic residence time within the ignition kernel that governs the overall ignition response. For different system pressures, the characteristic strain rates were weighted by density in order to isolate the chemical effects of pressure change from the physical effects of strain rate change [20].

3.2. Results and discussion

3.2.1. Comparison with previous study

The present experimental measurements were first compared with that of Seshadri et al. [6] in a similar stagnation-pool assembly at atmospheric pressure. Here we first note that the characteristic strain rate reported in Ref. [6] is the global strain rate, which is the ratio of the exit oxidizer velocity to the distance between the hot boundary and the stagnation plane. Furthermore, in Ref. [6] the measured temperature was corrected for radiative heat loss by assuming a spherical shape of the junction, a constant Nusselt number of 2.0, and a constant emissivity of 0.2, while in the present study we took the average of the temperatures corrected by spherical and cylindrical models, with the Nusselt number and emissivity calculated at each condition accordingly [21]. Thus in order to conduct a meaningful comparison of our results and those of Ref. [6], we have converted our measurements by applying the same definition of the strain rate as that of Ref. [6] and using the same procedure in post-processing the measured temperature. Figure 6 shows fairly close agreement between the two results, hence providing a consistency check for these two independent investigations.

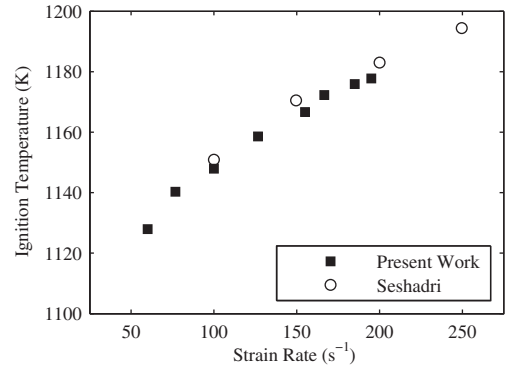

Fig. 6. Measurement of ignition temperature of present work and that of Ref. [6], 1 atm.

3.2.2. Comparison of measurements and simulation

Experimental measurements and numerical simulation of the ignition temperature were conducted at atmospheric pressure and slightly reduced and elevated pressures of 0.61 and 3.0 atmospheres, respectively, in order to capture the pressure effect on ignition. At 1 and 3 atmospheres the oxidizer was a mixture of 21% O_2 and 79% N_2, while at 0.61 atmospheres the O_2 concentration was increased to 30% to reduce the ignition temperature. The radiation-corrected ignition temperatures from measurement and the simulation results with the mechanism predictions are plotted versus density-weighted strain rates in Fig. 7, with error bars indicated.

It is first seen that the ignition temperatures calculated by the reduced mechanisms of Mech-LLL are lower than those of Mech-PRF by 40–70 K over the conditions examined, with the discrepancy increasing with pressure. As to the comparison between measurements and simulations, it is seen that while Mech-78 underpredicts the ignition temperature in the current non-premixed stagnation flow, Mech-PRF over predicts it. However, the simulations with both

kinetic models can be considered to provide fairly good agreement with the measurements in general, realizing that in previous studies the differences between simulations and measurements of non-premixed ignition temperature could easily exceed 100 K [9]. Figure 7 further shows that at 1 and 0.61 atmospheres, the two simulation curves are both within the error bars of the measurements, although the ignition temperatures calculated by Mech-78 are closer to measurements than those of Mech-PRF at 0.61 atmospheres. At 3 atmospheres Mech-PRF performs better than Mech-78, which yields the calculated ignition temperatures being systematically lower than the measurements by 50 K. Both models capture the aerodynamic straining effect well.

Finally it is of interest to note that the high-temperature model, Mech-PRF, is adequate to simulate the ignition response in the current non-premixed system, although it does not include the low-temperature pathways. This is possibly due to the relatively high ignition temperature (above 1000 K) in the present non-premixed stagnation flow, in which the diffusive and convective transport result in ignition temperatures much higher than those associated with the low-temperature chemistry alone.

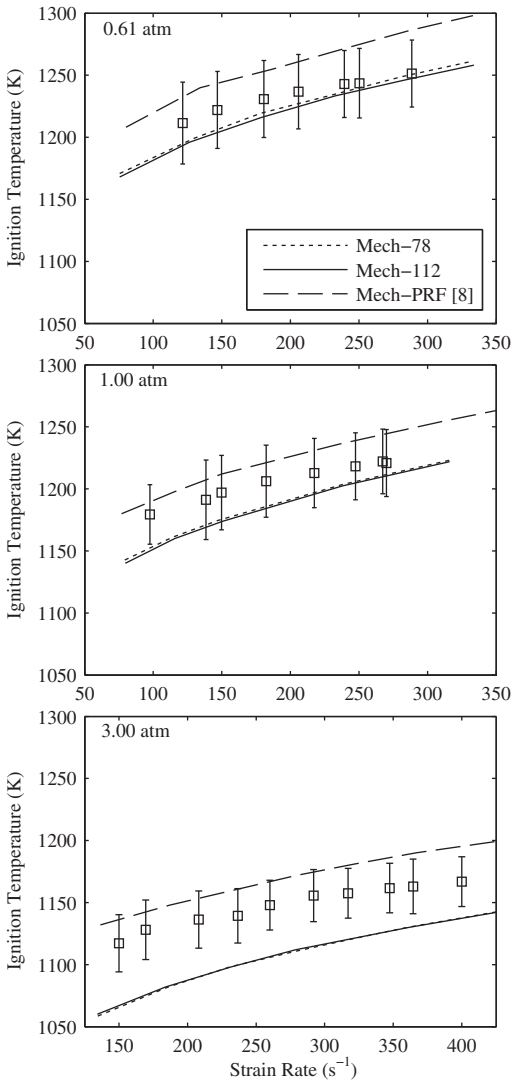

Fig. 7. Measurement and simulation of ignition temperature versus strain rate at various pressures.

3.2.3. Dominant chemistry

To identify the mechanism that controls ignition, a recently developed chemical explosive mode analysis (CEMA) [22,23], was conducted using Mech-112. In performing the CEMA, the explosion participation index (EPI) was employed to quantify the contributions of each elementary reaction to explosion, thus identifying the important reaction pathways to ignition.

The normalized EPIs evaluated at the ignition states for 0.61, 1 and 3 atmospheres are summarized in Figs. 8 and 9, at the same density-weighted strain rate of 300/s. The analysis shows that there are two groups of important reactions characterizing the *iso*-octane oxidation in the present non-premixed system. Specifically, the first group includes reactions involving the H_2/C_1 chemistry, as shown in Fig. 8. The major chain branching reactions, $H+O_2=O+OH$ and $CH_3+HO_2=CH_3O+OH$, promote ignition and have the largest EPIs for all the pressure conditions examined, demonstrating their significant roles in ignition. The major chain termination reactions, $H+O_2(+M)=HO_2(+M)$, $HO_2+OH=H_2O+O_2$ and $CH_3+CH_3(+M)=C_2H_6(+M)$, retard ignition as expected. These reactions have been identified to be important for methane as well as the larger hydrocarbons, since H and CH_3 are common radi-

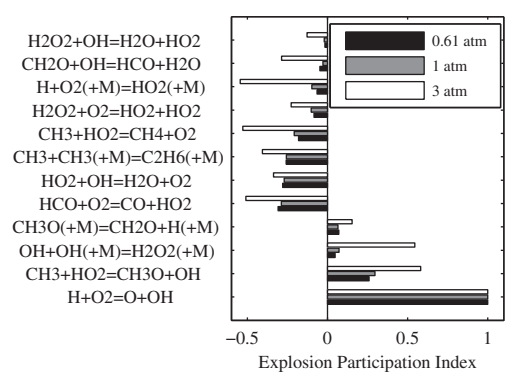

Fig. 8. Normalized explosion participation index at ignition for different pressures; $k_{local} = 300/s$.

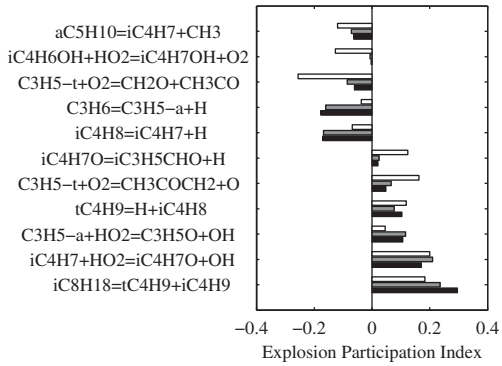

Fig. 9. Normalized explosion participation index at ignition for different pressures; $k_{local} = 300/s$.

cals for hydrocarbon oxidation. Furthermore, reactions involving the HO_2 chemistry become more significant as pressure increases.

The second group includes reactions that are more specific to *iso*-octane oxidation, as seen in Fig. 9. Specifically, the unimolecular decomposition of *iso*-octane to *tert*-butyl (tC_4H_9) and *iso*-butyl (iC_4H_9) plays a significant role in ignition, and its product further generates *iso*-butene (iC_4H_8) through unimolecular decomposition. Other important reactions include the subsequent oxidation of *iso*-butene, involving the radicals iC_4H_7, C_3H_6, C_3H_5-*a*, C_3H_5-*t*, etc. It is of interest to note that in a previous study of *iso*-octane in an atmospheric flow reactor at 1080 K [23], which is close to the temperature regime of interest in the present study, the oxidation of *iso*-octane was also found to produce primarily *iso*-butene (iC_4H_8) and propene (C_3H_6) as intermediate products, although the study was in a premixed configuration.

4. Concluding remarks

Experimental data of high accuracy were obtained for the laminar flame speeds and non-premixed stagnation-flow ignition temperatures of *iso*-octane at various pressures, with the laminar flame speeds measured using two independent techniques and showing close agreement. The measured laminar flame speeds at 1 atm pressure largely agree with the literature data, although at elevated pressures they are substantially lower than those from the only two sources of the literature data available. The non-premixed stagnation-flow measurements were shown to agree well with the measurements of Seshadri et al. [6], and yielded data of extended utility that are based on the local strain rate and with pressure variations.

The present experimental measurements were compared with two chemical kinetic mechanisms: the comprehensive mechanism of Curran et al. [7] (Mech-LLL) and the high-temperature mechanism of Chaos et al. [8] (Mech-PRF). In order to compare measurements with the mechanism of Curran et al., a skeletal (Mech-112) and a reduced (Mech-78) mechanism were first developed. Mech-LLL was found to yield significantly higher flame speeds from those predicted by Mech-PRF. A sensitivity-based analysis of the flame speed was conducted and the reactions responsible for the differences between the two mechanisms were identified, suggesting avenues for further scrutiny of the associated rate parameters. In comparing the experimental measurements of the non-premixed stagnation-flow ignition temperature against the two mechanisms, it was found that both mechanisms did a reasonably good job predicting the measurements, but both showed systematic differences with the measurements.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi: 10.1016/j.proci.2010.05.058.

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